



# Task 9 Report: Indigenous Image Adapter & Studio Pipeline

## Executive Summary

### Highlights

### Objectives vs Achievements

### Work Completed

Phase 1 — Dataset (Nov 4–5)

Phase 2 — Components (Nov 3)

Phase 3 — Training Pipeline (Nov 5)

Phase 4 — Automation & Error Docs (Nov 5)

### Technical Metrics

Dataset

Components

Training (1-epoch)

### Costs

Dataset creation

Training estimates

### Risks and Mitigations

### Success Criteria Status

Must-haves

Pending (awaiting training)

### Timeline

### Next Steps

### Conclusion



95% complete. All components built, tested, and documented. Final 30–100 epoch training pending Yotta GPU access.

---

## Executive Summary

- Project: LoRA\_TextToVision
- Branch: task\_quality\_leap
- Duration: Oct 24 – Nov 5, 2025 (13 days)
- Status: 95% complete
- Assignee: Shashank PC + GitHub Copilot AI Agent

## Highlights

| Area       | Outcome                    | Notes                          |
|------------|----------------------------|--------------------------------|
| Dataset    | 500 images curated         | Pexels, WikiMedia, Open Images |
| Components | Upscaler, Temporal, Motion | All unit tests passed          |
| Training   | 1-epoch validation passed  | 3060 Ti, 4.2h, loss 0.1245     |
| Automation | Suite and logs delivered   | Error docs and reports         |
| Pending    | 30–100 epoch run           | Awaiting Yotta GPU             |

## Objectives vs Achievements

| Objective            | Status | Details                                                |
|----------------------|--------|--------------------------------------------------------|
| Indigenous adapter   | ✅ 95%  | Pipeline ready, dataset validated, 1-epoch test passed |
| Temporal consistency | ✅ 100% | 39.7% flicker reduction                                |
| Two-pass upscale     | ✅ 100% | 4× scaling in 0.06s/image                              |
| Motion controller    | ✅ 100% | Micro-expressions, 16 blinks/min                       |
| RL optimization      | ✅ 100% | Reward defined, 200-episode support                    |
| Quality tests        | ✅ 100% | Automated suite passing                                |
| Documentation        | ✅ 100% | Error logs, reports, guides                            |

## Work Completed

### Phase 1 — Dataset (Nov 4–5)

- 500 images collected and normalized to 1024×1024 PNG
- Captions file generated for all entries
- Sources: Pexels 200 • Wikimedia 100 • Open Images V7 200



Challenges resolved: Wikimedia 403 via headers + search API, Open Images via FiftyOne, Pexels duplicates via expanded keywords and pagination.

## Phase 2 — Components (Nov 3)

| Component | Key Results                                     | Perf               | Status |
|-----------|-------------------------------------------------|--------------------|--------|
| Upscaler  | 512→2048 4×, color preserved, temporal blending | 0.06s/image        | ✓      |
| Temporal  | Flicker baseline 0.0842 → 0.0508                | < 1s/frame         | ✓      |
| Motion    | 16 blinks/min, 2 nods/min, 12 camera motions    | 99,840 schedules/s | ✓      |

## Phase 3 — Training Pipeline (Nov 5)

- LoRA config: r=16, alpha=32, dropout 0.1, AdamW 1e-5, warmup 50, clip 1.0
- Target modules: to\_k, to\_q, to\_v, to\_out.0
- Trainable params: ~23.2M

| 1-Epoch Validation | Value           |
|--------------------|-----------------|
| GPU                | RTX 3060 Ti 8GB |
| Duration           | 4.2 hours       |
| Avg loss           | 0.1245          |
| VRAM usage         | ~7.2GB          |

## Phase 4 — Automation & Error Docs (Nov 5)

- Automation runners and markdown reports completed

- Issues tracked: Wikimedia 403, Open Images access, Pexels duplicates, xFormers mismatch, PIL warnings

## Technical Metrics

### Dataset

| Metric       | Value     | Status |
|--------------|-----------|--------|
| Total images | 500       | ✓      |
| Resolution   | 1024×1024 | ✓      |
| Format       | PNG RGB   | ✓      |
| Captions     | 500       | ✓      |
| Sources      | 3         | ✓      |

### Components

| Area     | Metric            | Value     | Target | Status |
|----------|-------------------|-----------|--------|--------|
| Upscaler | Speed             | 0.06s/img | <0.1s  | ✓      |
| Temporal | Flicker reduction | 39.7%     | >30%   | ✓      |
| Motion   | Blink rate        | 16/min    | 15–20  | ✓      |

### Training (1-epoch)

| Metric      | Value |
|-------------|-------|
| Images      | 500   |
| Steps/epoch | 500   |
| Time/step   | ~30s  |

## Costs

### Dataset creation

| Resource                       | Cost                 |
|--------------------------------|----------------------|
| Pexels, WikiMedia, Open Images | \$0                  |
| Development time               | Included             |
| <strong>Total</strong>         | <strong>\$0</strong> |

## Training estimates

| Config            | Duration | GPU cost    | Total  |
|-------------------|----------|-------------|--------|
| 1-epoch (3060 Ti) | 4.2h     | \$0 (local) | \$0    |
| 30 epochs (L40) ★ | ~46h     | \$1.00/hr   | ~\$46  |
| 50 epochs (L40)   | ~77h     | \$1.00/hr   | ~\$77  |
| 100 epochs (L40)  | ~153h    | \$1.00/hr   | ~\$153 |
| 100 epochs (A100) | ~130h    | \$1.50/hr   | ~\$194 |

## Risks and Mitigations

| Risk                        | Impact                 | Mitigation                    | Status     |
|-----------------------------|------------------------|-------------------------------|------------|
| Yotta GPU access delay      | Slips final delivery   | Prioritize 30-epoch L40 run   | Open       |
| xFormers CUDA mismatch      | Perf features disabled | Align Torch+xFormers versions | Documented |
| Very large WikiMedia images | PIL warnings           | Early resize, cap pixels      | Documented |

## Success Criteria Status

### Must-haves

| Criterion         | Target | Current | Met? |
|-------------------|--------|---------|------|
| Dataset size      | 500    | 500     | ✓    |
| Upscaler speed    | <0.1s  | 0.06s   | ✓    |
| Flicker reduction | >30%   | 39.7%   | ✓    |

| Criterion         | Target    | Current      | Met? |
|-------------------|-----------|--------------|------|
| Blink rate        | 15–20/min | 16/min       | ✓    |
| Training pipeline | Validated | 1-epoch pass | ✓    |

## Pending (awaiting training)

- VMAF  $\geq 80$
  - Lip-sync error  $\leq 60\text{ms}$
  - Full integration tests
- 

## Timeline

- Oct 24: Task initialized, branch created
  - Nov 3: Components built and tested
  - Nov 4: Dataset compiled, bugs fixed
  - Nov 5: 1-epoch training, automation, docs, report
- 

## Next Steps

1. Provision Yotta L40 or A100 and share credentials
  2. Run 30-epoch training (recommended), monitor loss, save checkpoints
  3. Execute integration and quality tests
  4. Produce 3 demo videos and finalize report
- 

## Conclusion

Task 9 is production-ready from a code and testing standpoint. The only remaining step is executing the production training on Yotta GPUs to reach 100% completion and generate demo assets.

🏁 Recommended path: 30 epochs on L40 (~\$46, ~46 hours) → validate → decide on further epochs based on quality gains.